

Gargoyle Gecko (*Rhacodactylus auriculatus*)

Difficulty: Easy / Beginner

Gargoyle geckos are medium-sized New Caledonian geckos known for the bony horn-like ridges on the head, prehensile tail, and wide range of striped, blotched, and reticulated color patterns. Adults are typically about 8-10 inches long and commonly weigh around 45-65 grams. With good care, a lifespan of 15-20 years or more is realistic.

They are native to Grande Terre, New Caledonia, where they use humid forest edges and maquis shrubland. Field observations show that they are primarily nocturnal, often becoming most active during the first few hours after sunset, and that they forage on shrubs and small trees. Their wild diet is unusually broad for a gecko and includes arthropods, plant material, and occasional vertebrate prey. Captive care should therefore provide climbing structure, a day/night light cycle, a mild thermal gradient, regular insect prey, and a nutritionally complete gecko diet.

Shopping List

- Front-opening vertical enclosure, minimum 18" x 18" x 36" for a juvenile over roughly 12g or an adult; larger is preferred
- Secure screen or ventilated top with good airflow
- Low-wattage white/clear halogen or incandescent basking bulb
- Dimming thermostat or lamp dimmer for heat control
- Low-output linear UVB fixture suitable for Ferguson Zone 1
- Plug-in light timer
- Digital probe thermometer/hygrometer and infrared temperature gun
- Pressure sprayer or automatic misting system
- Reptile-safe water conditioner if using chlorinated tap water
- 2-4" tropical substrate; drainage layer and screen separator if bioactive
- Clean leaf litter and moisture-retentive planting material
- Sturdy cork bark, branches, vines, ledges, and dense live or artificial foliage
- Elevated feeding ledge with small food and water cups
- High-quality nutritionally complete crested/gargoyle gecko diet
- Appropriately sized crickets, roaches, black soldier fly larvae, silkworms, or similar feeder insects
- Calcium supplement for feeder insects; use little or no added D3 when appropriate UVB is provided
- Feeding tongs and a digital gram scale for routine weight checks

Housing

Gargoyle geckos are arboreal to semi-arboreal and need substantially more usable vertical space than the old 18" x 18" x 24" standard. For an established juvenile over about 12g or an adult, 18" x 18" x 36" should be treated as a minimum; 24" x 18" x 36", 24" x 24" x 36", or larger is better. Hatchlings can be started in a smaller, well-furnished grow-out enclosure so feeding and hydration are easy to monitor, but they should be moved up as they grow.

Fill the enclosure with usable structure rather than leaving a tall open box. Provide sturdy horizontal and diagonal branches, cork rounds, vines, ledges, and foliage at several heights. Gargoyle geckos use vertical space but are less adhesive and more heavy-bodied than crested geckos, so broad stable perches and multiple routes through the enclosure are especially useful. Aim for dense cover and visual barriers while still leaving enough open space for normal jumping and movement.

Provide a shallow water cup at all times even if the gecko usually drinks droplets after misting. Keep heavy decor stable and supported so it cannot shift if the gecko jumps or digs around the base.

Lighting & UVB

Gargoyle geckos are primarily nocturnal, but they still need a clear daytime light period. Use a roughly 12-hour day/night cycle, or gradually vary the photoperiod with the seasons. Bright daytime illumination from a plant LED can be used in planted enclosures, but the animal must always have shaded retreats.

Low-level UVB is strongly recommended. Field observations document occasional daytime exposure, and modern reptile-lighting standards place this type of nocturnal/shade-dwelling gecko in a low-UV range. A practical target is a UV Index around 1.0-2.0 at the upper basking branch, with dense foliage and lower areas providing complete shade. A low-output linear T5 such as an appropriately distanced ShadeDweller-style lamp works well. Mesh can reduce UVB substantially, so distance charts are only a starting point; a Solarmeter 6.5 is the best way to verify the actual UVI at the gecko.

Place the UVB on the same side as the basking heat so heat, visible light, and UVB form one optional basking zone. Do not place glass or plastic between the UVB lamp and the enclosure, because ordinary glass and most plastics block useful UVB. Do not use red, blue, purple, or black “night” bulbs; darkness at night is part of a normal circadian cycle.

Heating

Gargoyle geckos do best with access to a gentle thermal gradient rather than one uniform room temperature. Use a low-wattage overhead white/clear heat lamp on one side of the enclosure, controlled so that the rest of the enclosure remains substantially cooler. If your room already produces the correct gradient, extra heat may be unnecessary, but the enclosure should still be monitored rather than assumed to be safe.

Basking branch / surface: 82-84°F (28-29°C)

General daytime air: 72-80°F (22-27°C)

Cool lower zone: 70-75°F (21-24°C)

Nighttime: 65-72°F (18-22°C) is a normal safe drop for most healthy adults

Avoid sustained enclosure temperatures above about 85-86°F. New Caledonian geckos can overheat quickly in small or poorly ventilated enclosures. Use a digital probe thermometer for air temperature and an infrared thermometer for perch and basking-surface temperatures. A heat mat should not be the primary heat source for an arboreal adult enclosure.

Humidity

Humidity should cycle instead of staying constantly saturated. A useful daily pattern is roughly 50-70% during the drier part of the day, rising to about 80-100% immediately after a thorough evening mist, then gradually falling again as the enclosure ventilates. A light morning mist can be added if the enclosure dries too quickly.

The enclosure should smell fresh and the upper surfaces should have time to dry between mistings. Constantly wet glass, waterlogged substrate, stale air, or persistent heavy mold indicate too much water or too little ventilation. Mist foliage and climbing surfaces so the gecko can drink droplets, but still keep a clean water cup available at all times.

Substrate

For established juveniles and adults, use 2-4 inches of a moisture-retentive tropical substrate such as a pesticide- and fertilizer-free organic topsoil/coco mix, ReptiSoil-style blend, or ABG-type mix. Add clean leaf litter over the surface. Bioactive enclosures should include adequate drainage, a screen separator, deeper substrate, live plants, and a cleanup crew appropriate for the setup.

Paper towel is appropriate for quarantine, medical monitoring, or very small juveniles while feeding and stool output are being established. Avoid reptile carpet, pine or cedar, large bark chips, and other coarse particles that can catch toes or be difficult to pass if swallowed. Spot-clean waste and spoiled food promptly, and refresh or replace non-bioactive substrate as it becomes soiled rather than waiting for odor or mold problems.

Food

Gargoyle geckos are omnivores, and field studies show a higher proportion of animal prey than is typical for many other New Caledonian geckos. In captivity, use a reputable nutritionally complete crested/gargoyle gecko diet as the staple and provide live insects regularly for protein, hunting behavior, and dietary variety. Complete diets should be mixed according to the manufacturer's directions and replaced before they spoil or dry hard in the cup.

Hatchlings and juveniles: offer fresh complete gecko diet daily; offer appropriately sized insects about 2-3x/week

Adults: offer fresh complete gecko diet every other day or about 3x/week; offer insects about 1-2x/week

Useful feeders include dubia or discoid roaches, crickets, black soldier fly larvae, silkworms, and small hornworms. Rotate feeder species when possible and gut-load insects for 24-48 hours before feeding. Hard-bodied insects should generally be no wider than the space between the gecko's eyes.

Waxworms and other very fatty feeders should be treats rather than staples.

Wild gargoyle geckos sometimes consume small vertebrates, but vertebrate prey is not necessary for a healthy captive diet and should not be used as a routine food. Complete gecko diet plus appropriately supplemented insects is easier to balance and carries less risk of excessive calories. Baby food and plain fruit should never replace a complete gecko diet; fresh fruit can be an occasional enrichment treat only.

Supplements

Do not add extra calcium or vitamins directly to a nutritionally complete gecko diet unless a reptile veterinarian has instructed you to do so. The prepared diet is already formulated to be balanced. Live feeder insects, however, should be lightly dusted because most feeder insects have an unfavorable calcium-to-phosphorus ratio.

When properly measured UVB is provided, use plain calcium or a low-D3 calcium product according to its label. If UVB is not available, a calcium product containing D3 becomes more important, but relying only on supplements is less physiologically flexible than allowing the gecko access to low-level UVB and shade. Avoid stacking multiple vitamin products, which can lead to excess fat-soluble vitamins. Adult females can lay infertile eggs, so maintaining adequate calcium intake and body condition is particularly important.

Decor

Decor is functional habitat, not decoration alone. Use cork bark, sturdy branches, vines, ledges, broad leaves, and plants to create covered routes from the bottom to the top of the enclosure. Include several sleeping retreats and perches at different temperatures and light levels so the gecko can choose where to rest. Horizontal and diagonal branches are especially important because they provide stable footing for this relatively heavy-bodied species.

Handling Tip: Give a newly acquired gecko at least 1-2 weeks to settle and eat reliably before routine handling. Let the gecko walk onto your hands rather than grabbing it from above, keep sessions short at first, and handle over a low safe surface. Gargoyle geckos can drop the tail when frightened, although unlike crested geckos they are capable of regrowing it.

Cohabitation

Single housing is the safest default for gargoyle geckos. Field observations suggest that individuals partition perch height, likely in part to reduce aggressive encounters, and captive gargoyles can bite, tail-nip, compete for food, or injure smaller geckos. Never house two males together and never mix substantially different sizes; larger gargoyle geckos are capable of preying on smaller lizards, including other geckos.

For normal pet care, house one gargoyle gecko per enclosure. Breeding introductions should be intentional and closely monitored, with enough space and duplicate resources to separate animals immediately if aggression, weight loss, persistent hiding, or tail damage occurs.